TECHNICAL ANALYSIS

Nine Companion Problems — EMA, RSI and Stochastic Oscillator

Securities Analysis and Portfolio Management

Solution + Full Working Notes for Every Problem

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Exponential Moving Average

PROBLEM 1 *(EXPONENTIAL MOVING AVERAGE — Problem 1 of 3)*

Ishaan tracks the 50-day EMA on Vindhya Power Ltd. as his desk's standard trend filter, the same 50-day convention used across every section of the analyst-training program. The 50-day EMA, computed from the average of the preceding 50 sessions, currently stands at ₹2,145.80. Today's closing price is ₹2,189.50, and Ishaan needs the updated 50-day EMA before the desk's evening trend review begins.

Questions:

i. Compute the EMA multiplier (exponent) for a 50-day EMA.

ii. Compute the updated 50-day EMA.

SOLUTION

Formula:

EMA (today) = EMA (prior) + [(Close − EMA (prior)) × Exponent]

Exponent = 2 ÷ (N + 1), where N = EMA period

Working:

Step 1: Exponent = 2 ÷ (50 + 1) = 2 ÷ 51 = 0.0392.

Step 2: Updated EMA = ₹2,145.80 + [(₹2,189.50 − ₹2,145.80) × 0.0392] = ₹2,145.80 + [₹43.70 × 0.0392] = ₹2,145.80 + ₹1.71 = **₹2,147.51**.

WORKING NOTES — FULL STEP-BY-STEP CALCULATIONS

The table below repeats the exact five-column layout used to build up an EMA — Day, Close, Prior EMA, Difference, and the resulting new EMA — so every intermediate figure is visible, not just the final answer.

Day	Close (₹)	Prior EMA (₹)	Difference (Close − Prior)	Diff × Exponent (0.0392)	New EMA
Today	2,189.50	2,145.80	43.70	1.71	2,147.5

PROBLEM 2 *(EXPONENTIAL MOVING AVERAGE — Problem 2 of 3)*

Meera is rolling Chambal Agro Ltd.'s 50-day EMA forward over two fresh sessions before her weekly trend note goes out, applying the same rolling procedure the desk uses every single day, regardless of how many sessions have passed since the last update. The 50-day EMA currently stands at ₹845.60. The stock then closes at ₹862.30 today and ₹858.90 tomorrow, and Meera needs both EMA values updated in sequence.

Questions:

i. Compute today's updated 50-day EMA.

ii. Using today's EMA as the new base, compute tomorrow's updated 50-day EMA.

SOLUTION

Formula:

EMA(today) = EMA (prior) + [(Close − EMA(prior)) × Exponent]

Exponent = 2 ÷ (N + 1), where N = EMA period

Working:

Step 1: Exponent = 2 ÷ 51 = 0.0392 (unchanged for both steps, since N = 50 throughout).

Step 2: Today's EMA = ₹845.60 + [(₹862.30 − ₹845.60) × 0.0392] = ₹845.60 + [₹16.70 × 0.0392] = ₹845.60 + ₹0.65 = **₹846.25**.

Step 3: Tomorrow's EMA = ₹846.25 + [(₹858.90 − ₹846.25) × 0.0392] = ₹846.25 + [₹12.65 × 0.0392] = ₹846.25 + ₹0.50 = **₹846.75**.

WORKING NOTES — FULL STEP-BY-STEP CALCULATIONS

Each row below is one full application of the EMA update — note how tomorrow's row reuses today's finished EMA as its own 'Prior EMA' input, exactly the way the recursive formula is meant to chain forward.

Day	Close (₹)	Prior EMA (₹)	Difference	Diff × 0.0392	New EMA (₹)
Today	862.30	845.60	16.70	0.65	846.25
Tomorrow	858.90	846.25	12.65	0.50	846.75

PROBLEM 3 *(EXPONENTIAL MOVING AVERAGE — Problem 3 of 3)*

Arjun's fund runs a three-session rolling review of Ganga Shipping Ltd.'s 50-day EMA every Friday, chaining the update forward one session at a time rather than recomputing from scratch. This week's 50-day EMA opens at ₹1,320.00. Over the next three sessions, the stock closes at ₹1,345.80, then ₹1,338.20, then ₹1,352.60, and Arjun needs all three updated EMA values, each carried forward from the previous one.

Questions:

i. Compute the updated 50-day EMA after each of the three sessions in turn.

ii. State the final EMA after all three updates.

SOLUTION

Formula:

EMA(today) = EMA(prior) + [(Close − EMA(prior)) × Exponent]

Exponent = 2 ÷ (N + 1), where N = EMA period

Working:

Step 1: Exponent = 2 ÷ 51 = 0.0392 (same value for all three sessions, since N = 50 throughout).

Step 2: Session 1: ₹1,320.00 + [(₹1,345.80 − ₹1,320.00) × 0.0392] = ₹1,320.00 + ₹1.01 = **₹1,321.01**.

Step 3: Session 2: ₹1,321.01 + [(₹1,338.20 − ₹1,321.01) × 0.0392] = ₹1,321.01 + ₹0.67 = **₹1,321.69**.

Step 4: Session 3: ₹1,321.69 + [(₹1,352.60 − ₹1,321.69) × 0.0392] = ₹1,321.69 + ₹1.21 = **₹1,322.90**.

WORKING NOTES — FULL STEP-BY-STEP CALCULATIONS

Three full rows this time — each session's finished EMA becomes the very next session's 'Prior EMA', so a single arithmetic slip in Session 1 would carry through both later sessions. That chaining is exactly what real EMA tracking looks like day after day.

Session	Close (₹)	Prior EMA (₹)	Difference	Diff × 0.0392	New EMA (₹)
1	1,345.80	1,320.00	25.80	1.01	1,321.01
2	1,338.20	1,321.01	17.19	0.67	1,321.69
3	1,352.60	1,321.69	30.91	1.21	1,322.90

PROBLEM 2 ***(MOVING AVERAGE CONVERGENCE DIVERGENCE — Problem 2 of 3)***

Priya is reviewing the MACD of **Krishna Engineering Ltd.** as of **20 August** and wants to trace how the indicator changed over the two preceding sessions. On **17 August**, the stock's 12-day EMA was ₹1,205.00 and its 26-day EMA was ₹1,198.00. The signal line immediately before 17 August was ₹5.50. On **18 August**, the 12-day EMA increased to ₹1,210.00 and the 26-day EMA increased to ₹1,201.00. Priya needs the MACD and signal-line values for both sessions, calculated sequentially.

Questions:

i. Compute the MACD for 17 August and 18 August.

ii. Using the signal line immediately before 17 August, compute the signal line for 17 August.

iii. Using the 17 August signal line as the new base, compute the signal line for 18 August.

iv. Compute the MACD histogram for both sessions and comment on the change in momentum.

SOLUTION

Formula:

MACD = 12-day EMA − 26-day EMA

Signal Line (today) = Signal Line (prior) + [(MACD − Signal Line (prior)) × Exponent]

Exponent = 2 ÷ (N + 1), where N = signal-line period

Histogram = MACD − Signal Line

Working:

Step 1: Signal-line exponent = 2 ÷ (9 + 1) = **0.20**.

Step 2: 17 August MACD = 1,205.00 − 1,198.00 = **₹7.00**.

Step 3: 17 August Signal Line = 5.50 + [(7.00 − 5.50) × 0.20] = 5.50 + 0.30 = **₹5.80**.

Step 4: 17 August Histogram = 7.00 − 5.80 = **₹1.20**.

Step 5: 18 August MACD = 1,210.00 − 1,201.00 = **₹9.00**.

Step 6: 18 August Signal Line = 5.80 + [(9.00 − 5.80) × 0.20] = 5.80 + 0.64 = **₹6.44**.

Step 7: 18 August Histogram = 9.00 − 6.44 = **₹2.56**.

Step 8: The MACD increased from ₹7.00 to ₹9.00, while the histogram increased from ₹1.20 to ₹2.56. This indicates that **positive momentum strengthened between 17 and 18 August**.

WORKING NOTES — FULL STEP-BY-STEP CALCULATIONS

Each row represents one complete MACD update. Notice that the 18 August signal-line calculation uses the **finished 17 August signal line as its prior value**, demonstrating the recursive nature of the signal line.

Date	12-day EMA (₹)	26-day EMA (₹)	MACD (₹)	Prior Signal (₹)	Signal Line (₹)	Histogram (₹)
17 August	1,205.00	1,198.00	7.00	5.50	5.80	1.20
18 August	1,210.00	1,201.00	9.00	5.80	6.44	2.56

PROBLEM 3 ***(MOVING AVERAGE CONVERGENCE DIVERGENCE — Problem 3 of 3)***

Sanjay is analysing **Narmada Consumer Products Ltd.** as of **20 August** to determine whether the stock's recent momentum is strengthening. The following 12-day and 26-day EMA values are available for the three preceding sessions. The signal line immediately before 17 August was ₹7.20.

Date	12-day EMA (₹)	26-day EMA (₹)
17 August	1,108.00	1,101.20
18 August	1,112.00	1,104.20
19 August	1,116.00	1,107.60

Sanjay wants to determine whether the MACD has crossed above its signal line and whether the momentum signal has strengthened.

Questions:

i. Compute the MACD for each of the three sessions.

ii. Using the previous signal line of ₹7.20, calculate the signal line for 17 August, 18 August, and 19 August sequentially.

iii. Calculate the MACD histogram for each session.

iv. Identify whether a bullish MACD crossover occurred. If so, on which date?

v. What does the crossover and the movement of the histogram suggest about the stock's momentum?

SOLUTION

Formula:

MACD = 12-day EMA − 26-day EMA

Signal Line (today) = Signal Line (prior) + [(MACD − Signal Line (prior)) × Exponent]

Exponent = 2 ÷ (N + 1), where N = signal-line period

Histogram = MACD − Signal Line

Working:

Step 1: Signal-line exponent = 2 ÷ (9 + 1) = **0.20**.

Step 2: 17 August MACD = 1,108.00 − 1,101.20 = **₹6.80**.

Step 3: 17 August Signal Line = 7.20 + [(6.80 − 7.20) × 0.20] = 7.20 − 0.08 = **₹7.12**.

Step 4: 17 August Histogram = 6.80 − 7.12 = **−₹0.32**.

Step 5: 18 August MACD = 1,112.00 − 1,104.20 = **₹7.80**.

Step 6: 18 August Signal Line = 7.12 + [(7.80 − 7.12) × 0.20] = 7.12 + 0.14 = **₹7.26**.

Step 7: 18 August Histogram = 7.80 − 7.26 = **₹0.54**.

Step 8: 19 August MACD = 1,116.00 − 1,107.60 = **₹8.40**.

Step 9: 19 August Signal Line = 7.26 + [(8.40 − 7.26) × 0.20] = 7.26 + 0.23 = **₹7.49**.

Step 10: 19 August Histogram = 8.40 − 7.49 = **₹0.91**.

Step 11: On 17 August, the MACD (₹6.80) was below the signal line (₹7.12). On 18 August, the MACD (₹7.80) moved above the signal line (₹7.26). Therefore, a **bullish MACD crossover occurred on 18 August**.

Step 12: The histogram changed from −₹0.32 on 17 August to +₹0.54 on 18 August and then increased to +₹0.91 on 19 August. This indicates that **positive momentum not only emerged but strengthened after the bullish crossover**.

The crossover is a **potential bullish momentum signal**, but it does not guarantee that the stock price will rise. It should be considered alongside the broader trend and other technical evidence.

WORKING NOTES — FULL STEP-BY-STEP CALCULATIONS

The table below shows the complete three-session sequence. The key point is the transition from a **negative histogram to a positive histogram**, which makes the crossover visible numerically rather than relying only on a chart.

Date	12-day EMA (₹)	26-day EMA (₹)	MACD (₹)	Prior Signal (₹)	Signal Line (₹)	Histogram (₹)
17 August	1,108.00	1,101.20	6.80	7.20	7.12	−0.32
18 August	1,112.00	1,104.20	7.80	7.12	7.26	+0.54
19 August	1,116.00	1,107.60	8.40	7.26	7.49	+0.91

Relative Strength Index (RSI)

PROBLEM 4 *(RELATIVE STRENGTH INDEX — Problem 1 of 3)*

Kavita is computing the standard 14-day RSI on Tungabhadra Jewels Ltd. for the shared cross-section model every faculty member uses, so her work must follow the textbook procedure exactly. Over the last 15 sessions, the stock has closed at ₹340, ₹344, ₹342, ₹347, ₹349, ₹346, ₹350, ₹353, ₹351, ₹355, ₹358, ₹356, ₹360, ₹357, and ₹362, and Kavita needs the final RSI reading before her weekly momentum log is due.

Questions:

i. Separate the fourteen daily changes into gains and losses and compare the average gain and average loss.

ii. Compute RS and the final RSI and classify the reading.

SOLUTION

Formula:

RS = Average Gain ÷ Average Loss

RSI = 100 − [100 ÷ (1 + RS)]

Working:

Step 1: Total Gains = 34; Average Gain = 34 ÷ 14 = 2.43.

Step 2: Total Losses = 12; Average Loss = 12 ÷ 14 = 0.86.

Step 3: RS = 2.43 ÷ 0.86 = 2.83.

Step 4: RSI = 100 − [100 ÷ (1 + 2.83)] = 100 − 26.09 = **73.91**.

Step 5: Since RSI (73.91) > 70, the reading sits in overbought territory.

WORKING NOTES — FULL STEP-BY-STEP CALCULATIONS

Every one of the fourteen daily changes, classified as a gain or a loss exactly the way the textbook table does it — nothing here is skipped or pre-summed.

Day	Close (₹)	Change	Gain	Loss
1	340	—	—	—
2	344	+4	4	0
3	342	−2	0	2
4	347	+5	5	0
5	349	+2	2	0
6	346	−3	0	3
7	350	+4	4	0

8	353	+3	3	0
9	351	−2	0	2
10	355	+4	4	0
11	358	+3	3	0
12	356	−2	0	2
13	360	+4	4	0
14	357	−3	0	3
15	362	+5	5	0

PROBLEM 5 *(RELATIVE STRENGTH INDEX — Problem 2 of 3)*

Rahul is running the same standard 14-day RSI model on Sutlej Sugar Ltd. after a strong run in stock and wants to check exactly how extreme the reading has become before recommending fresh buying to his clients. Over the last 15 sessions, the stock has closed at ₹500, ₹506, ₹510, ₹508, ₹514, ₹519, ₹517, ₹522, ₹527, ₹525, ₹530, ₹534, ₹532, ₹538, and ₹543, and Rahul needs the completed RSI table before the desk closes today.

Questions:

i. Separate the fourteen daily changes into gains and losses, and compare the average gain and average loss.

ii. Compute RS and the final RSI, and state whether it warrants caution on fresh buying.

SOLUTION

Formula:

RS = Average Gain ÷ Average Loss

RSI = 100 − [100 ÷ (1 + RS)]

Working:

Step 1: Total Gains = 51; Average Gain = 51 ÷ 14 = 3.64.

Step 2: Total Losses = 8; Average Loss = 8 ÷ 14 = 0.57.

Step 3: RS = 3.64 ÷ 0.57 = 6.38.

Step 4: RSI = 100 − [100 ÷ (1 + 6.38)] = 100 − 13.56 = **86.44**.

Step 5: RSI (86.44) is deep into overbought territory — Rahul should treat fresh buying with real caution, though this alone does not prove the trend is about to reverse.

WORKING NOTES — FULL STEP-BY-STEP CALCULATIONS

The full fourteen-day classification, day by day — notice how few losses this series has, which is exactly what pushes RSI this high.

Day	Close (₹)	Change	Gain	Loss
1	500	—	—	—
2	506	+6	6	0
3	510	+4	4	0
4	508	−2	0	2
5	514	+6	6	0
6	519	+5	5	0
7	517	−2	0	2
8	522	+5	5	0

9	527	+5	5	0
10	525	−2	0	2
11	530	+5	5	0
12	534	+4	4	0
13	532	−2	0	2
14	538	+6	6	0
15	543	+5	5	0

PROBLEM 6 *(RELATIVE STRENGTH INDEX — Problem 3 of 3)*

Ananya is computing the same 14-day RSI on Brahmaputra Telecom Ltd. after weeks of the stock going essentially nowhere and wants a precise number rather than relying on a purely visual impression of the choppy chart. Over the last 15 sessions, the stock has closed at ₹220, ₹223, ₹219, ₹222, ₹218, ₹221, ₹224, ₹220, ₹223, ₹219, ₹222, ₹225, ₹221, ₹224, and ₹220, and Ananya needs the finished RSI reading for her weekly momentum log.

Questions:

i. Separate the fourteen daily changes into gains and losses and compute the average gain and average loss.

ii. Compute RS and the final RSI, and comment on what this reading suggests for a swing trade.

SOLUTION

Formula:

RS = Average Gain ÷ Average Loss

RSI = 100 − [100 ÷ (1 + RS)]

Working:

Step 1: Total Gains = 24; Average Gain = 24 ÷ 14 = 1.71.

Step 2: Total Losses = 24; Average Loss = 24 ÷ 14 = 1.71.

Step 3: RS = 1.71 ÷ 1.71 = 1.00.

Step 4: RSI = 100 − [100 ÷ (1 + 1.00)] = 100 − 50.00 = **50.00**.

Step 5: An RSI of exactly 50, alongside a narrow, choppy price range, indicates no directional momentum edge in either direction — not a basis for a fresh swing trade in either direction.

WORKING NOTES — FULL STEP-BY-STEP CALCULATIONS

Gains and losses come out almost perfectly balanced across all fourteen days — this is what a genuinely rangebound stock's RSI table looks like underneath the surface.

Day	Close (₹)	Change	Gain	Loss
1	220	—	—	—
2	223	+3	3	0
3	219	−4	0	4
4	222	+3	3	0
5	218	−4	0	4

6	221	+3	3	0
7	224	+3	3	0
8	220	−4	0	4
9	223	+3	3	0
10	219	−4	0	4
11	222	+3	3	0
12	225	+3	3	0
13	221	−4	0	4
14	224	+3	3	0
15	220	−4	0	4

Stochastic Oscillator

PROBLEM 7 *(STOCHASTIC OSCILLATOR — Problem 1 of 3)*

Vikram is computing the standard 14-day stochastic oscillator on Narmada Paints Ltd. for the shared momentum dashboard, every section reports against. Over the last 14 sessions, the highest high is ₹845, and the lowest low is ₹792. Today's close comes in at ₹818, and the two most recent prior %K readings on his sheet were 62.5 and 68.0, giving him a short trend to extend before the desk meeting.

Questions:

i. Compute today's %K.

ii. Compute today's %D as the three-period average of %K, and comment on the reading.

SOLUTION

Formula:

%K = [(Close − Lowest Low) ÷ (Highest High − Lowest Low)] × 100

%D = 3-period Moving Average of %K

Working:

Step 1: %K (today) = [(₹818 − ₹792) ÷ (₹845 − ₹792)] × 100 = (26 ÷ 53) × 100 = **49.06**.

Step 2: %D (today) = (62.5 + 68.0 + 49.06) ÷ 3 = 179.56 ÷ 3 = **59.85**.

Step 3: Both readings sit comfortably between the 20 and 80 thresholds — momentum is roughly neutral, with no overbought or oversold extreme.

WORKING NOTES — FULL STEP-BY-STEP CALCULATIONS

The %K formula broken into its three separate pieces — the range, the numerator, and the final ratio — followed by the %D addition spelled out term by term.

Quantity	Calculation	Result
Range (HH − LL)	845 − 792	53
Numerator (Close − LL)	818 − 792	26
Ratio	26 ÷ 53	0.4906
%K	0.4906 × 100	49.06
%D sum	62.5 + 68.0 + 49.06	179.56
%D	179.56 ÷ 3	59.85

PROBLEM 8 *(STOCHASTIC OSCILLATOR — Problem 2 of 3)*

Pooja is completing the same 14-day stochastic computation on Kaveri Chemicals Ltd. after a sharp rally and wants to check whether the reading has moved into caution territory before recommending fresh buying to her clients. Over the last 14 sessions, the highest high is ₹612, and the lowest low is ₹558. Today's close is ₹601, and the two most recent prior %K readings were 79.5 and 85.0.

Questions:

i. Compute today's %K.

ii. Compute today's %D, and state whether both readings now sit in overbought territory.

SOLUTION

Formula:

%K = [(Close − Lowest Low) ÷ (Highest High − Lowest Low)] × 100

%D = 3-period Moving Average of %K

Working:

Step 1: %K (today) = [(₹601 − ₹558) ÷ (₹612 − ₹558)] × 100 = (43 ÷ 54) × 100 = **79.63**.

Step 2: %D (today) = (79.5 + 85.0 + 79.63) ÷ 3 = 244.13 ÷ 3 = **81.38**.

Step 3: %D (81.38) sits above the conventional 80 overbought threshold, though today's own %K (79.63) is just under it — Pooja should treat fresh buying with caution rather than chase the rally further.

WORKING NOTES — FULL STEP-BY-STEP CALCULATIONS

Same three-piece breakdown as before, so the borderline call on today's %K versus the clearly overbought %D is easy to see in the raw numbers.

Quantity	Calculation	Result
Range (HH − LL)	612 − 558	54
Numerator (Close − LL)	601 − 558	43
Ratio	43 ÷ 54	0.7963
%K	0.7963 × 100	79.63
%D sum	79.5 + 85.0 + 79.63	244.13
%D	244.13 ÷ 3	81.38

PROBLEM 9 *(STOCHASTIC OSCILLATOR — Problem 3 of 3)*

Lakshmi tracks Coromandel Foods Ltd.'s 14-day stochastic across two consecutive sessions for her fund's rolling momentum sheet, since the desk wants to see how %K moves as the 14-day window itself shifts forward by one day. On Day N, the trailing 14-day high is ₹438 and the low is ₹402, with a close of ₹429. One session later, on Day N+1, a fresh session high pushes the trailing high to ₹441 while the low is unchanged at ₹402, with a close of ₹433. The prior %K reading before Day N was 70.2.

Questions:

i. Compute %K for Day N and for Day N+1.

ii. Compute %D as of Day N+1, using the prior reading, Day N, and Day N+1.

SOLUTION

Formula:

%K = [(Close − Lowest Low) ÷ (Highest High − Lowest Low)] × 100

%D = 3-period Moving Average of %K

Working:

Step 1: Day N: %K = [(₹429 − ₹402) ÷ (₹438 − ₹402)] × 100 = (27 ÷ 36) × 100 = **75.00**.

Step 2: Day N+1: %K = [(₹433 − ₹402) ÷ (₹441 − ₹402)] × 100 = (31 ÷ 39) × 100 = **79.49**.

Step 3: %D (Day N+1) = (70.2 + 75.00 + 79.49) ÷ 3 = 224.69 ÷ 3 = **74.90**.

Step 4: %K is rising session over session as the trailing high itself rises — the rolling 14-day window is doing real work here, not just the day's own close.

WORKING NOTES — FULL STEP-BY-STEP CALCULATIONS

Both sessions' %K broken into the same three pieces as before, placed side by side so the effect of the trailing high moving from ₹438 to ₹441 is visible directly in the denominator.

Quantity	Day N	Day N+1
Highest High (14)	438	441
Lowest Low (14)	402	402
Close	429	433
Range (HH − LL)	36	39
Numerator (Close − LL)	27	31
%K	75.00	79.49